

PAPER CODE	EXAMINER	DEPARTMENT	TEL
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1st SEMESTER 2024/25 Individual Coursework

BSc Information Management &
Information Systems

Advanced Database Management

INSTRUCTIONS TO CANDIDATES

1. A single copy of your work must be submitted electronically through Learning Mall
Online by **11:59 pm, Friday, 20 Dec. 2024** (China Standard Time, UTC/ GMT+8).
2. This coursework will count for 70% of the final mark in this course.

Coursework Title:

Individual coursework: Implementing a Database using Microsoft Access

Scenario:

The Scenario is the same as in the GROUP COURSEWORK.

Task Details/Description:

The primary task for this part of the coursework is to **individually produce a fully tested and functional database application that has been implemented using Microsoft Access.** Please note, you MUST at the minimum have the following functionalities in your database application to score a passing mark for the corresponding grading component. Please refer to the pages at the end of this document for the grading components and criteria.

- The database *must* support several business processes described in the scenario context and specification. You may change some of the design produced in the group coursework based on the feedback received from the instructors.
- Include *functionality (forms, queries, reports and charts)* to support the main operations of the company. The database should be able to: **add, edit and remove books, order books, generate invoices, reorder low stock and produce useful management information** (e.g. total value of stocks).
- *Entities should be identified by a concise and relevant set of attributes.* They must include the basic attributes and be realistic but need not be fully comprehensive for the prototype.

Continue/revise on what has been produced in GROUP coursework

- Include **at least two non-trivial queries.** Your queries should demonstrate the use of multiple tables, parameters, calculated fields and multi-stage query. **You should also produce appropriate SQL code for at least one of your queries.** The code should be annotated in the report to demonstrate your understanding (please refer to ID2.4 below in the deliverables section).
- Include **at least one unbounded form, one trivial and one non-trivial form, that will help to realise key business needs**, i.e. gather inputs and produce the correct output, which are user-friendly and displayed in an appropriate manner to the user (e.g. home page – unbounded form, add a new customer – trivial form, add new customer order, reorder books – non-trivial forms).
- Include **two neatly formatted Access reports** to provide useful management information. Include at least one level of grouping, and at least one summary/calculated field (e.g. customer invoice, most valued customer based on purchase orders, most popular books sold or reordered, total expenses incurred towards each publisher, most valued employee based on sales).
- Include **one Access chart** that illustrates some key operational information from the application to the organisation (e.g. sales made each month, popular books).

For this coursework, you may continue working with the deliverables (ERM, Tables and Sample

records) produced in the group coursework, after making appropriate changes considering the summative feedback. Including any extra features, you may wish to – adding value to the application in functionality and usability may earn extra credit. Your report must clearly state these features.

Deliverables:

You MUST submit the following two deliverables (ID1 and ID2) to complete this coursework.

ID1: Fully tested and functional database that fulfils the functional requirements meeting the business needs specified in the scenario.

ID2: Report not exceeding 1500 words (+/-10%), which MUST include the following. The front cover page, table of contents, headers, sub-headers, figure and tables, labels as well as annotations, citations and references, are not included in the word count (i.e. 1500 words). Estimated word limits for each section are provided below to help you write the report and have some idea of how much to focus on each section, while deciding the contents. You must use Harvard referencing style for your references and citations. **Please note, your report MUST include the following at a (or the) minimum to score a passing mark for the corresponding grading component. Please refer to the last two pages of this document for the grading components and criteria.**

- **ID2.1:** The **front cover page** of the report must include your student number and title of the assignment [*No word count applies*]
- **ID2.2:** Simple instructions briefly describing **how to operate the database**. The instructions must be written as if for a typical employee with limited knowledge of Microsoft Access. You should use figures and annotate them to better articulate the instructions. [*estimated word limit – 200 words*]
- **ID2.3:** Brief description of the **most significant features in the database**, for example, how these features meet the needs of the business, addressing the current limitations, explained with the aid of screenshots from your database application (as applicable). Justify the features and/or design decisions citing appropriate literature. You may like to focus on **two-three most significant features** that will help to realise the value of your application to the business. [*estimated word limit – 400 words*]
- **ID2.4:** Include a brief description of **two non-trivial queries**. In your description, state why they are non-trivial and how they meet the needs of the organisation. Additionally, please include figures/screenshots to show each query [table-view] you have generated and the corresponding output. You must also include the SQL code for one of these queries, and briefly explain the code in your own words, demonstrating your understanding, how the query will execute. [*estimated word limit – 300 words, the SQL code will not be included in the word count*]
- **ID2.5:** Include a brief description of **two management reports**. In your description, state why the reports are useful for the management, and how they will add value to the business. Please include figures to show sample report outputs you have generated. You may annotate them (as applicable). [*estimated word limit – 150 words*]
- **ID2.6:** Include a brief description of **one non-trivial chart**. In your description, clearly

explain the value of the chart to the organisation. Please include a figure to show the sample output. You may annotate them (as applicable). [*estimated word limit – 100 words*]

- **ID2.7:** Include a brief description of how you have **tested the database** to ensure it is working properly, capable of handling errors, and generating the correct output for your queries. Please include a couple of screenshots to provide evidence of testing. [*estimated word limit – 150 words*]
- **ID2.8:** A brief **reflection** on the strengths (at least *two*) and weaknesses (at least *two*) of your application. You must also suggest at least *two* ways in which you could significantly improve or enhance the application. This section should also include a brief discussion of your own performance and your learning. [*estimated word limit – 200 words*]

Warning! If your database cannot be opened, is corrupt or is otherwise inaccessible, or your media contains malware, you will score zero for this part of the assignment.

Please ensure you have created one or more back-up copies of your work in suitable places using appropriate mediums available to you, to avoid losing the work you may have already done.

Your work will be assessed on robustness, functionality meeting the business requirements, usability and the overall quality of your design. The working and fully functional database will contribute to the majority weight for this coursework, hence please create necessary back-ups.

Feedback Methods:

Formative feedback will be available before the final submission, to help you develop and refine your work. This will include the following methods:

- **Weekly labs** – these will include tasks and activities (related to the coursework) to help you apply knowledge from the lectures, develop your own solution and obtain feedback in session to help you to improve your work. You are therefore encouraged to engage during the lectures and seminars to make progress, receive feedback, clarify doubts, and refine your work.
- **Assignment clinic** – during weeks 11 and 12, there will sessions dedicated to review the assessment requirements, clarify areas of doubt and obtain further feedback on your work.
- **Office hours** – You are advised to use office hours early in the term, before the submission deadline, so you have time to act on the feedback.
- **Discussion Forum in LM** – you can ask questions, discuss among peers and receive responses via the discussion forum. Turnaround time for a response from the module tutor is 2-3 working days or it will be given in the following lab.

Summative feedback will be provided after the final submission. This accompanies your final grade and uses the assessment criteria shown at the end of this document.

Submission:

You are required to submit two files: (1) Your Microsoft Access Application; (2) Your Report.

The coursework is going to be submitted electronically. Further instructions about how to do this will be provided during a workshop session.

Assessment Weighting for the Module:

The weight of the individual coursework is **70%**

Assessment Criteria

Marking guidelines are provided at the end of this document.

Contact

If you have any questions regarding this coursework, please email to ou.liu@xjtlu.edu.cn or Wenzhe.Liang19@student.xjtlu.edu.cn, OR you may book office hours with the teaching team.

Individual Coursework marking Guidelines

	70% and above	60-69	50-59	40-49	Less than 40%
Overall database (30%) [accompanying submission – Microsoft Access database file]	Application addresses all the requirements and considers additional non-trivial needs which have the potential to add business value. All the functionalities execute without any errors. Database uses advance features including some beyond those taught in the class. Database is effectively presented to a professional standard and is very well-structured. The application demonstrates excellent and very effective use of technology to create, and modify databases, using relevant queries, and produce management reports.	Application addresses most requirements in the specification. Most functionalities execute properly without any errors. Use of advance features including some beyond those taught in the class is limited. Database is effectively presented and well-structured. The application demonstrates very good, and effective use of the technology to create, and modify databases, using relevant queries, and produce management reports.	Application addresses some requirements in the specification. Some functionalities execute properly without any errors. The presentation is clear and has a suitable structure. The application demonstrates good use of the technology to create, and modify databases, using relevant queries, and produce management reports.	Application addresses few requirements in the specification. Several functionalities do not execute properly, as intended. Database lacks suitable presentation and structure. The application demonstrates limited use of the technology to create, and modify databases, using relevant queries, and produce management reports.	Application addresses very few requirements in the specification. Most functionalities do not execute properly and are not devoid of errors. Database is incomplete, and presentation is unclear and disorganised. The application demonstrates very limited use of the technology to create, and modify databases, using relevant queries, and produce management reports.
Forms (15%) [accompanying submission – forms in the Microsoft Access database file]	All the forms and sub-forms support realistic tasks meeting and exceeding the business needs. One or more forms support non-trivial tasks beyond the specification, which have the potential to add value. There are no redundant/unused/incomplete forms. The design is professional and there is evidence of employing design principles. All forms execute without any	Most forms and sub-forms support realistic tasks meeting most of the business needs. Almost none of the forms in the application are redundant/unused/incomplete. The form design is effective and there is little evidence of employing design principles. Most forms execute as intended without any errors.	Some forms have been designed to support trivial tasks to meet some of the business needs. There are some redundant/unused/incomplete forms. Form design is generic, and there is no evidence of employing design principles. Some forms execute without any errors.	Few forms have been designed to support very trivial tasks meeting few of the business needs. Many forms are redundant/unused/incomplete. Limited attention has been paid to design usable forms. Few forms execute as intended without any errors.	Most forms do not meet the business needs and are incomplete/redundant. Most forms execute with errors.

	errors.				
Report and Chart (15%) [accompanying submission – reports and charts in the Microsoft Access database file and their description in the report]	All the reports and charts provide useful management information meeting and exceeding business needs. Reports and charts look professional, and execute without any errors, displaying the correct output. One or more reports/chart provide non-trivial information beyond the specification, considering future needs. There are no redundant/unused/incomplete reports and charts.	Most reports and charts provide useful management information meeting most of the business needs. Most reports and charts are well designed, and execute without any errors, displaying the correct output. There are almost no redundant/unused/incomplete reports and charts.	Some reports and charts have been designed to provide trivial information meeting some of the business needs. Some reports and charts execute without any errors, displaying the correct output. There are some redundant/unused/incomplete reports and charts.	Few reports and charts have been designed to provide very trivial information meeting few of the business needs. Few of them execute without any errors, displaying the correct output. Many reports and charts are redundant/unused/incomplete.	Most reports and charts do not meet the business needs and are incomplete/redundant. Most reports and charts execute with errors.
	70% and above	60-69	50-59	40-49	Less than 40%
Queries (20%) [accompanying submission – queries in the Microsoft Access database file and description of queries in the report]	All queries display very useful and meaningful information to the management in a suitable format and execute correctly without any errors. The non-trivial queries have used all the following techniques: multiple tables, parameters, conditions, calculated fields and a multi-stage execution. One or more queries address requirements	Most queries display useful information to the management in a suitable format and execute correctly without any errors. The non-trivial queries have used at least three of the following techniques: multiple tables, parameters, conditions, calculated fields and a multi-stage execution. There are almost no redundant/unused/incomplete queries.	Some queries have been implemented to display information for the management. These queries execute with limited errors. The non-trivial queries have used at least two of the following techniques: multiple tables, parameters, conditions, calculated fields and a multi-stage execution. There are some redundant/unused/incomplete	Few queries have been implemented to display information for the management, and most of them are not free from errors. The non-trivial queries have used one of the following techniques: multiple tables, parameters, conditions, calculated fields and a multi-stage execution. There are several redundant/unused/incomplete queries.	Couple of queries have been implemented to display some information, and none of them are free from errors. The non-trivial queries have used none of the following techniques: multiple tables, parameters, conditions, calculated fields and a multi-stage execution. Most queries are redundant/unused/incomplete.

	beyond the specification, considering the future needs of the organisation. There are no redundant/unused/incomplete queries.		queries.		
Report (20%) [accompanying submission – report submitted as a word or PDF document]	Very well structured and presented. Effectively and convincingly communicates to a professional standard. Writing is very clear and concise, flows logically, leaves no room for reduction and contains no (or only minor) spelling/grammatical errors. A wide range of supporting evidence including images, charts and diagrams is clearly, logically and professionally presented. Harvard referencing standard is consistently and appropriately applied throughout. Report demonstrates excellent understanding and appreciation of the role of database in meeting the need for information management in business organizations	Well-structured and presented that communicates effectively. Writing is clear and concise and is largely free of spelling/grammatical errors. A range of supporting evidence including images, charts and diagrams is clearly, and appropriately presented. Harvard referencing standard is appropriately applied throughout with few inconsistencies. Report demonstrates very good understanding and appreciation of the role of database in meeting the need for information management in business organizations	Communication is generally clear but may lack a logical structure. Writing contains spelling/grammatical errors that affect clarity. Supporting evidence is used but may be inappropriate or inadequate in some instances. Harvard referencing standard is applied but may contain some errors or inconsistencies. Report demonstrates some understanding and appreciation of the role of database in meeting the need for information management in business organizations	A lack of clear structure and spelling/grammatical errors make the work difficult to read, but some logic is evident. There is some use of supporting evidence, but this may be limited or inappropriate. Referencing is used but with several errors or inconsistencies. Report demonstrates limited understanding and appreciation of the role of database in meeting the need for information management in business organizations	Very disorganised and unclear. Writing is unprofessional or contradictory and does not coherently explain the writer's logic. Supporting evidence is inappropriate, and unclear. Sources are not referenced, or referencing is inappropriate and inconsistent. Report demonstrates very limited understanding and appreciation of the role of database in meeting the need for information management in business organizations